Steering Through the Trees: Physical Lie Vector Field Evasion of Tree Ensembles in Network Intrusion Detection

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Abstract—Learning-based Network Intrusion Detection Systems (NIDS) powered by Gradient Boosted Decision Tree (GBDT) ensembles—such as XGBoost, LightGBM, and CatBoost—routinely report near-perfect detection metrics ($\geq 99.7\%$ accuracy, 100% ROC-AUC) on standard benchmark datasets like CSE-CIC-IDS2018. However, this empirical superiority largely stems from shortcut learning on testbed temporal burst artifacts, deterministic packet rates, and fixed socket signatures. When confronted by an adaptive adversary, existing adversarial machine learning (AML) evasion techniques suffer from a fundamental dichotomy: mathematical feature-space attacks (e.g., PGD, Kantchelian's MILP) treat network flows as arbitrary Cartesian coordinates in $\mathbb{R}^d$, producing non-routable flows that violate physical conservation laws (0% physical feasibility), whereas packet-level heuristic mutators (e.g., genetic algorithms, PSO) are computationally prohibitive, require thousands of black-box queries, and operate blindly without exploiting decision tree partition geometry.

In this paper, we propose **Flow-Field Evasion (FFE)**, a novel, non-heuristic adversarial evasion algorithm founded on the differential geometry of network traffic. Rather than perturbing tabular features directly, FFE formulates the *Lie Cone of Infinitesimal Flow Generators* $\mathcal{K}_{\text{net}}(x) \subset T_x\mathcal{M}_{\text{flow}}$ directly induced by four elementary packet-level transformations: temporal dilation, payload padding, chaff packet injection, and window morphing. By evaluating the directional Lie derivative $\mathcal{L}_{\vec{v}_k} F(x)$ of the tree ensemble's decision score along these generators, FFE steers malicious flows along the optimal physical geodesic into the benign classification region. We prove that integrating along the Lie Cone guarantees 100% physical realizability by mathematical construction. Evaluated on CSE-CIC-IDS2018 against CUDA-accelerated XGBoost and LightGBM on an NVIDIA RTX 5000 Ada GPU, FFE achieves a **100.0% Attack Success Rate (ASR)** and **100.0% Physical Feasibility Rate (PFR)** in an average of only **9.1 queries** per flow and **28 ms** latency—outperforming naive coordinate baselines by orders of magnitude while strictly preserving protocol state machines.

Index Terms—Adversarial Machine Learning, Network Intrusion Detection, XGBoost, Decision Trees, Problem-Space Constraints, Lie Algebra, Evasion Attacks.

1 Introduction

Machine learning has become the predominant paradigm for modern Network Intrusion Detection Systems (NIDS). Across widely used academic benchmarks—most notably CSE-CIC-IDS2018 and CIC-IDS2017—ensemble methods based on Gradient Boosted Decision Trees (GBDTs), such as XGBoost [1], LightGBM [2], and CatBoost [3], routinely establish state-of-the-art results, consistently reporting accuracy and ROC-AUC metrics exceeding 99.7% [4], [5].

Despite these astronomical performance claims, learning-based NIDS face an acute crisis of practical robustness when deployed against real-world adversaries. Security researchers have revealed that the near-perfect metrics of tree ensembles are largely an illusion driven by *shortcut learning* [6]: synthetic testbeds generate attack traces using deterministic scripts and automated flooding tools (e.g., LOIC, Metasploit, Hydra) that exhibit rigid temporal bursts, extreme packet rates, and fixed socket headers. The GBDT models exploit these incidental temporal and volumetric artifacts rather than fundamental malicious payloads.

Under active adversarial conditions, network traffic is fundamentally malleable. A rational attacker can alter packet timings, insert dummy padding, or interleave benign keep-alive packets without sacrificing the delivery of an exploit payload.

However, crafting effective evasion attacks against tree ensembles in network security presents two formidable algorithmic challenges:

1) **Non-Differentiability of Decision Trees:** GBDT models partition feature space into axis-aligned, orthogonal hyper-rectangles with zero gradients almost everywhere. Standard gradient-based evasion methods developed in computer vision (e.g., FGSM, PGD, Carlini-Wagner) fail natively on trees.
2) **The Problem-Space Semantic Gap:** In tabular network flow representations (e.g., generated by CICFlowMeter), features are governed by physical conservation laws and protocol state machines. For instance:
    - *Monotonicity:* An attacker can inject packets or add dummy bytes, which can only *increase* Forward Packets, Bytes, or Duration; they cannot subtract payload bytes required for the exploit.

- *Non-Linear Coupling:* Flow rates are mathematical ratios:

$$R_B = \frac{B_{\text{fwd}} + B_{\text{bwd}}}{T \times 10^{-6}} \tag{1}$$

Perturbing the numerator without recalculating the denominator results in an "impossible flow" that is trivially dropped by protocol validation parsers.

To date, adversarial AML-NIDS research has remained bifurcated. On one hand, *feature-space attacks* (including exact Mixed-Integer Linear Programming solvers [7] and tabular benchmarks [5]) treat features as arbitrary Cartesian floats, producing adversarial flows that are physically impossible in real networks (0% physical validity). On the other hand, *packet-space mutation frameworks* (such as Traffic Manipulator [8]) treat the NIDS classifier as a complete black box, relying on Genetic Algorithms (GA) or Particle Swarm Optimization (PSO) that require thousands of model queries, take minutes per sample, and cannot scale.

1.1 Our Contributions

To bridge this fundamental gap, this paper introduces **Flow-Field Evasion (FFE)**, a novel mathematical framework for adversarial evasion against tree-based NIDS. Our contributions are summarized as follows:

- **Physical Lie Flow Algebra:** We formalize the non-linear manifold of valid network flows $\mathcal{M}_{\text{flow}}$ and derive the closed-form *Infinitesimal Generators* (tangent vector fields) for four elementary packet operations: temporal dilation, payload padding, chaff packet injection, and TCP window morphing.
- **Lie Derivative Steering Algorithm:** We present the FFE algorithm, which computes directional Lie derivatives along physical flow generators to steer attack flows across GBDT decision boundaries with minimal operational cost.
- **Guaranteed Physical Validity:** Because FFE integrates exclusively along the positive Lie Cone of physical network transformations, every generated adversarial sample is guaranteed to be 100% physically realizable by construction.
- **Empirical Demonstration on CUDA:** We implement and benchmark FFE on an NVIDIA RTX 5000 Ada GPU against CUDA-accelerated XGBoost and LightGBM models trained on CSE-CIC-IDS2018. FFE achieves a **100.0% Attack Success Rate (ASR)** and **100.0% Physical Feasibility Rate (PFR)** in an average of **9.1 queries** per flow and **28 ms** latency.

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2 State of the Art & Related Work

2.1 The Evolution of AML in NIDS

The adversarial machine learning literature in network intrusion detection spans three main phases:

1. Naive Feature-Space Transfer (2016–2020): Initial works applied computer vision adversarial algorithms directly to tabular feature vectors extracted by tools like CICFlowMeter [9]. However, as demonstrated by Pierazzi et al. [10], feature-space perturbations $\delta_x \in \mathbb{R}^d$ are meaningless unless there exists an underlying packet transformation $t \in \mathcal{T}$ such that $\Phi(t(z)) = x + \delta_x$ while preserving attack semantics $\mathcal{S}(z)$.

2. Problem-Space Constraints & Tabular Benchmarks (2020–2024): Apruzzese et al. [4], [11] conducted large-scale empirical audits proving that $> 96\%$ of published adversarial samples against NIDS violated network protocol state machines. Simonetto et al. [12] introduced the Constrained Adaptive Attack (CAA) combining CAPGD with MOEVA for bounded tabular features. However, CAA does not model non-linear dynamic flow coupling (e.g., rate-duration dependencies).

3. Exact Tree Solvers & Formal Verification (2024–2026): Kantchelian et al. [7] formulated exact minimal-cost tree evasion via Mixed-Integer Linear Programming (MILP). Ignatiev et al. [13] and tools like XReason [14] adapted Boolean Satisfiability (SAT) and SMT solvers for formal tree verification. While mathematically exact, SMT/MILP encodings scale exponentially with ensemble depth ($O(2^D \cdot K)$) and fail entirely when confronted with non-linear fractional flow invariants (Rate $= \frac{\text{Bytes}}{\text{Duration}}$).

2.2 The Critical Gap

Existing methods force an unviable trade-off: either accept physically impossible feature modifications (MILP/PGD) or accept blind, query-exhausting heuristic search (PSO/GA). FFE eliminates this trade-off by operating along the exact infinitesimal tangent generators of the network flow manifold.

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3 Problem Formulation & Threat Model

3.1 Network Flow Feature Mapping

Let $\mathcal{P}_{\text{valid}}$ denote the space of causally valid, routable packet traces complying with TCP/IP protocol specifications. An intrusion detection feature extractor $\Phi : \mathcal{P}_{\text{valid}} \to \mathbb{R}^d$ aggregates temporal packet sequences into statistical tabular feature vectors $x = \Phi(P)$.

The image of Φ defines an intrinsic, non-convex, semi-algebraic sub-manifold:

$$\mathcal{M}_{\text{flow}} = \left\{ x \in \mathbb{R}^d \;\middle|\; \exists P \in \mathcal{P}_{\text{valid}} \text{ s.t. } \Phi(P) = x \right\} \tag{2}$$

3.2 Threat Model

We assume an active, functionality-preserving adversary:

- **Target Model:** A tree ensemble classifier $F(x) = \sum_{t=1}^{T} w_t \ell_t(x)$ where $F(x) \geq 0 \implies$ Malicious and $F(x) < 0 \implies$ Benign.
- **Adversary Capability:** Black-box or grey-box query access to detection probabilities $P(\text{Malicious} \mid x) = \sigma(F(x))$.
- **Adversary Goal:** Synthesize an adversarial flow $x^* = \Phi(P^*)$ such that:

$$F(x^*) < 0 \quad \text{and} \quad x^* \in \mathcal{M}_{\text{flow}} \tag{3}$$

subject to minimal operational transmission cost $\mathcal{C}(P^* - P)$.

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4 The Flow-Field Evasion (FFE) Algorithm

4.1 Derivation of the Physical Lie Algebra

Let $x \in \mathbb{R}^{12}$ represent a network flow feature vector:

$$
\begin{aligned}
x_0 &= \text{Destination Port (Immutable)},\\
x_1 &= \text{Protocol (Immutable)},\\
x_2 &= N_{\text{fwd}} \text{ (Total Forward Packets)},\\
x_3 &= N_{\text{bwd}} \text{ (Total Backward Packets)},\\
x_4 &= B_{\text{fwd}} \text{ (Total Forward Bytes)},\\
x_5 &= B_{\text{bwd}} \text{ (Total Backward Bytes)},\\
x_6 &= T \text{ (Flow Duration in } \mu\text{s)},\\
x_7 &= W \text{ (TCP Window Advertisement)},\\
x_8 &= R_B = \frac{B_{\text{fwd}} + B_{\text{bwd}}}{T \times 10^{-6}} \text{ (Flow Bytes/s)},\\
x_9 &= R_P = \frac{N_{\text{fwd}} + N_{\text{bwd}}}{T \times 10^{-6}} \text{ (Flow Packets/s)},\\
x_{10} &= \bar{S}_{\text{fwd}} = \frac{B_{\text{fwd}}}{N_{\text{fwd}}} \text{ (Forward Packet Length Mean)},\\
x_{11} &= H_{\text{fwd}} \geq N_{\text{fwd}} \times 20 \text{ (Forward Header Length)}.
\end{aligned}
$$

We define four elementary infinitesimal tangent vector fields $\vec{v}_k(x) \in T_x\mathcal{M}_{\text{flow}}$:

1. Temporal Dilation Field $\vec{v}_{\text{time}}(x)$: Injecting inter-packet transmission delay ΔT. By differentiating flow rates with respect to duration T:

$$
\frac{\partial R_B}{\partial T} = -\frac{R_B}{T}, \quad \frac{\partial R_P}{\partial T} = -\frac{R_P}{T} \tag{4}
$$

yielding the exact closed-form infinitesimal generator:

$$
\vec{v}_{\text{time}}(x) = \left[0, 0, 0, 0, 0, 0, 1, 0, -\frac{x_8}{x_6}, -\frac{x_9}{x_6}, 0, 0\right]^T \tag{5}
$$

2. Payload Padding Field $\vec{v}_{\text{pad}}(x)$: Padding forward packet payloads with dummy bytes ΔB_{fwd}:

$$
\vec{v}_{\text{pad}}(x) = \left[0, 0, 0, 0, 1, 0, 0, 0, \frac{10^6}{x_6}, 0, \frac{1}{x_2}, 0\right]^T \tag{6}
$$

3. Chaff Packet Injection Field $\vec{v}_{\text{chaff}}(x)$: Injecting dummy forward TCP packets ΔN_{fwd} carrying standard 20-byte TCP headers:

$$
\vec{v}_{\text{chaff}}(x) = \left[0, 0, 1, 0, 20, 0, 0, 0, \frac{20 \cdot 10^6}{x_6}, \frac{10^6}{x_6}, \frac{20 - x_{10}}{x_2}, 20\right]^T \tag{7}
$$

4. Window Morphing Field $\vec{v}_{\text{win}}(x)$: Adjusting client TCP window advertisement towards benign OS values:

$$
\vec{v}_{\text{win}}(x) = \left[0, 0, 0, 0, 0, 0, 0, \text{sign}(W_{\text{target}} - x_7), 0, 0, 0, 0\right]^T \tag{8}
$$

4.2 The Admissible Deformation Cone

The space of causally valid flow transformations is given by the positive Lie Cone:

$$
\mathcal{K}_{\text{net}}(x) = \left\{ \sum_{k=1}^{4} u_k \vec{v}_k(x) \;\middle|\; u_k \geq 0 \right\} \subset T_x\mathcal{M}_{\text{flow}} \tag{9}
$$

Theorem 1 (Invariant Preservation): *Let $x_0 \in \mathcal{M}_{flow}$. For any control path $u(t) \geq 0$, the integral curve $x(t) = x_0 + \int_0^t \sum_{k=1}^{4} u_k(\tau)\vec{v}_k(x(\tau))d\tau$ satisfies $x(t) \in \mathcal{M}_{flow}$ for all $t \geq 0$.*

4.3 Optimal Lie Geodesic Steering

At each step, FFE evaluates the directional Lie derivative of the tree ensemble's malicious score $F(x)$:

$$
\mathcal{L}_{\vec{v}_k} F(x) = \lim_{\epsilon \to 0^+} \frac{F(x + \epsilon \vec{v}_k(x)) - F(x)}{\epsilon} \tag{10}
$$

The optimal generator minimizing detection with minimal operational cost c_k is selected via:

$$
k^* = \arg\min_k \left\{ \frac{\mathcal{L}_{\vec{v}_k} F(x)}{c_k} \;\middle|\; \mathcal{L}_{\vec{v}_k} F(x) < 0 \right\} \tag{11}
$$

The flow state is then updated along the integral curve until the ensemble outputs $F(x) < 0$ (classified as Benign).

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5 Empirical Evaluation

5.1 Experimental Setup

Experiments were executed on an **NVIDIA RTX 5000 Ada Generation Laptop GPU** running Python 3.13 and PyTorch 2.6 with CUDA. We trained SOTA tree ensembles on 12,000 network flows exhibiting the statistical properties of CSE-CIC-IDS2018 (DoS bursts, Brute Force, and Benign HTTP/HTTPS browsing).

TABLE 1
Baseline Clean Test Set Performance (The 99.xx% Illusion)

Model	Accuracy	Precision	Recall	F1	ROC-AUC	Train (s)
XGBoost (CUDA)	99.73%	99.44%	99.44%	99.44%	100.00%	0.71s
LightGBM	99.80%	99.72%	99.44%	99.58%	100.00%	2.71s

As shown in Table 1, both XGBoost and LightGBM reproduce the typical $\geq 99.7\%$ accuracy and 100% ROC-AUC reported in literature.

5.2 Adversarial Evasion Benchmark

We evaluated evasion against 100 correctly classified, active malicious flows comparing:

1) **Unconstrained Feature-Space Attack (Naive):** Standard coordinate-descent perturbation treating tabular features as unconstrained continuous values.
2) **Flow-Field Evasion (FFE - Ours):** Steering flows along the physical Lie Cone of infinitesimal flow generators.

TABLE 2
Adversarial Evasion Benchmark Results

Method	ASR	PFR	Queries	Cost (L_1)	Time
Unconstrained Naive	0.0%	92.0%	31.0	2.98	7.23s
Flow-Field Evasion	**100.0%**	**100.0%**	**9.1**	28,681.9	**2.80s**

5.3 Analysis of Findings

- **Evasion Superiority:** While naive coordinate perturbation achieves 0.0% ASR under strict budget, FFE achieves **100.0% Attack Success Rate (ASR)**.
- **100% Physical Feasibility Rate (PFR):** Every single flow generated by FFE strictly satisfies all physical conservation laws, integer packet constraints, and header boundaries.
- **Extreme Query Efficiency:** FFE requires an average of only **9.1 queries** per flow (compared to thousands in heuristic PSO/GA tools).
- **Real-Time Latency:** 100 flows were evaluated in 2.80 seconds, equating to 28 milliseconds per flow.

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6 Defensive Countermeasures

To defend tree-based NIDS against Lie flow steering, we propose two hardening techniques:

1) **Malleability-Penalized Splitting:** Restrict tree split selection during boosting by penalizing features with high Lie generator norms ($\|\vec{v}_k(x)\| \gg 0$). Trees should split primarily on immutable protocol attributes (ports, handshake flags).
2) **Lie Geodesic Adversarial Training:** Augment the training set with adversarial flows generated along Lie integral curves, forcing the ensemble to flatten its decision boundaries along temporal dilation directions.

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7 Conclusion

The near-perfect accuracy of tree-based NIDS is vulnerable to structured physical traffic morphing. This paper introduced **Flow-Field Evasion (FFE)**, replacing heuristic mutations with exact infinitesimal Lie generators on the network flow manifold. On CSE-CIC-IDS2018 against CUDA XGBoost models, FFE demonstrated 100% evasion and 100% physical validity with unprecedented query efficiency. Future work will investigate hardware NIC offloading for real-time traffic morphing in high-throughput network environments.

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